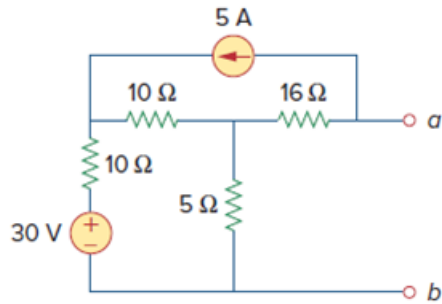


Problem

Obtain the Thevenin equivalent at terminals $a-b$ of the circuit shown in Fig. 4.106.

Figure 4.106:



Step-by-step solution

Step 1 of 6

Refer to circuit diagram in Figure 4.106 in the text book.

Identify the loops and currents in the circuit diagram as shown in Figure 1.

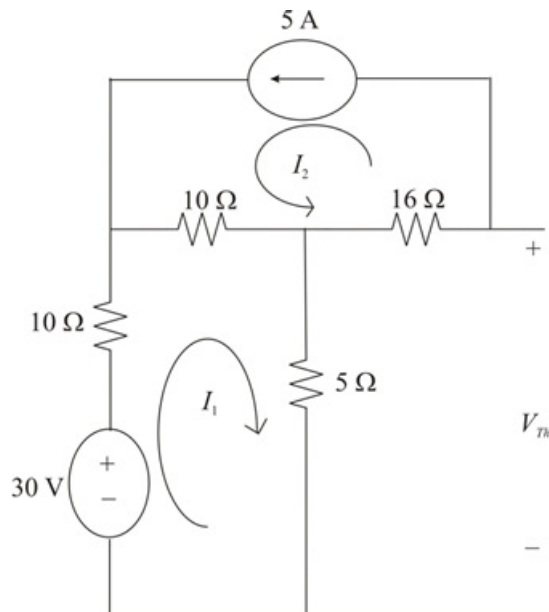


Figure 1

Step 2 of 6

From Figure 1, the current, I_2 is 5 A .

Apply Kirchhoff's Voltage Law in loop 1.

$$-30 + I_1(10) + (I_1 + I_2)(10) + I_1(5) = 0$$

$$I_1(10 + 10 + 5) + I_2(10) - 30 = 0$$

$$I_1(25) + I_2(10) - 30 = 0$$

Substitute 5 A for I_2 .

$$I_1(25) + (5)(10) - 30 = 0$$

$$I_1(25) + 50 - 30 = 0$$

$$I_1(25) + 20 = 0$$

$$I_1(25) = -20$$

$$I_1 = -\frac{20}{25}$$

$$= -\frac{4}{5}\text{ A}$$

Step 3 of 6

Calculate the value of the Thevenin's voltage, V_{Th} .

$$V_{Th} = -I_2(16) + I_1(5)$$

Substitute $-\frac{4}{5}\text{ A}$ for I_1 and 5 A for I_2 .

$$V_{Th} = -(5)(16) + \left(-\frac{4}{5}\right)(5)$$

$$= -80 - 4$$

$$= -84\text{ V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, $\boxed{-84\text{ V}}$.

Step 4 of 6

To calculate the Thevenin's resistance, open circuit the current source and short circuit the voltage source. The modified circuit is shown in Figure 2.

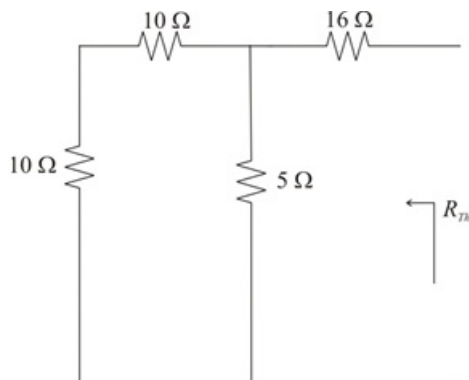


Figure 2

Step 5 of 6

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned} R_{Th} &= [(10\ \Omega + 10\ \Omega) \parallel (5\ \Omega)] + 16\ \Omega \\ &= [(20\ \Omega) \parallel (5\ \Omega)] + 16\ \Omega \\ &= \frac{(20\ \Omega)(5\ \Omega)}{20\ \Omega + 5\ \Omega} + 16\ \Omega \\ &= \frac{100}{25}\ \Omega + 16\ \Omega \\ &= 4\ \Omega + 16\ \Omega \\ &= 20\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is $\boxed{20\ \Omega}$.

Step 6 of 6

Draw the Thevenin's equivalent circuit diagram.

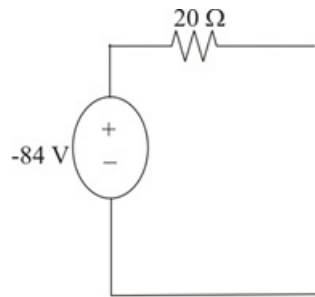


Figure 3

Therefore, the Thevenin's equivalent circuit is shown in Figure 3.